

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN DRAINAGE SYSTEM

MODULE - 1

TYPES OF DRAINAGE PATTERN

INDIAN GEOGRAPHY

PLAN OF ACTION



1. INDIAN CLIMATE
2. INDIAN SOILS
3. INDIAN NATURAL VEGETATION
4. INDIAN PHYSIOGRAPHY
 - HIMALAYAN RANGES
 - NORTHERN PLAINS
 - PENINSULAR PLATEAU
 - COASTAL PLAINS OF INDIA
 - ISLANDS OF INDIA
 - HILL RANGES OF THE PENINSULAR PLATEAU



INDIAN GEOGRAPHY

PLAN OF ACTION

5. INDIAN DRAINAGE SYSTEM

- INDUS RIVER SYSTEM
- GANGA RIVER SYSTEM
- BRAHMAPUTRA RIVER SYSTEM
- PENINSULAR RIVER SYSTEM



INDIAN GEOGRAPHY

PLAN OF ACTION

6. MINERAL DISTRIBUTION IN INDIA

- IRON-ORE DISTRIBUTION
- COAL FORMATION AND DISTRIBUTION
- PETROLEUM & ITS DISTRIBUTION
- MANGANESE ORE & ITS DISTRIBUTION
- DISTRIBUTION OF GOLD & SILVER
- COPPER, NICKEL & CHROMITE
- NON-FERROUS MINERALS DISTRIBUTION
- DISTRIBUTION OF NUCLEAR MINERALS
- DISTRIBUTION OF DIAMOND & GRAPHITE
- NON-METALLIC MINERALS IN INDIA



INDIAN DRAINAGE SYSTEM

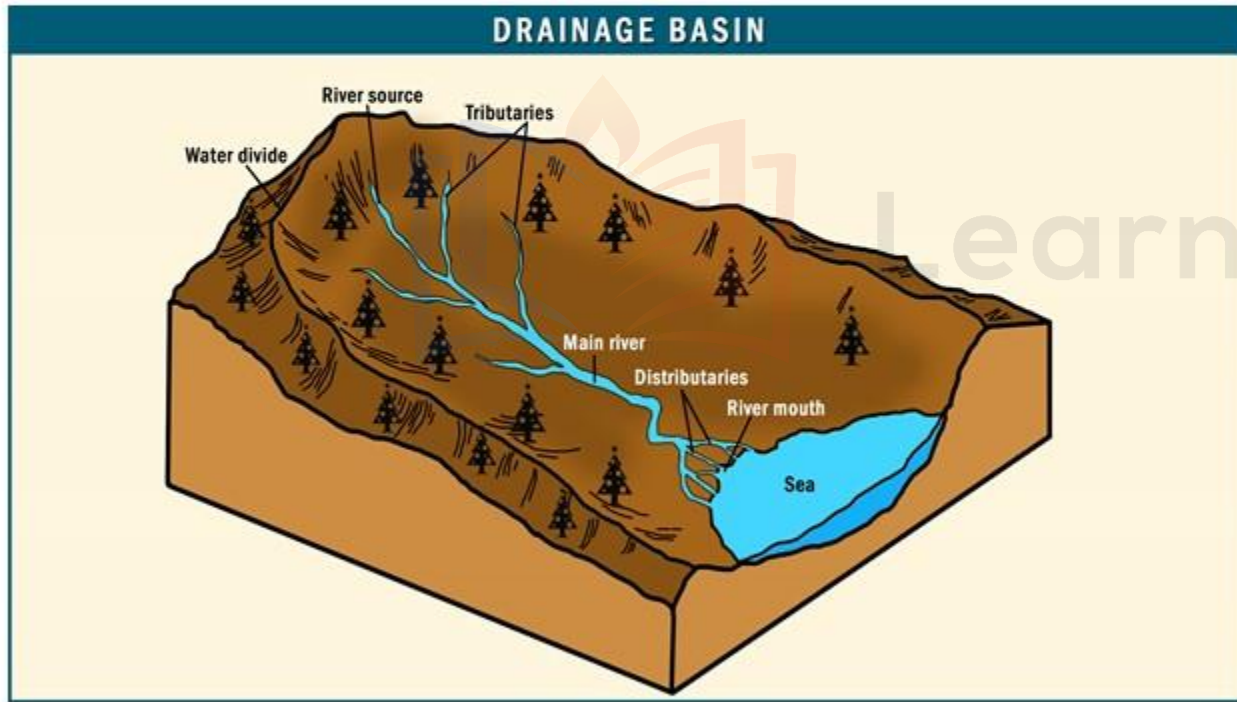
WHAT DO YOU MEAN BY A DRAINAGE SYSTEM?



- 1 The drainage system is an **integrated system of tributaries of a river**, which collect and funnel surface water to the sea, lake or some other body of water.
- 2 In other words, the flow of water through **well-defined channels** is known as drainage and **the network of such channels** is known as drainage system.

INDIAN DRAINAGE SYSTEM

WHAT DO YOU MEAN BY A DRAINAGE BASIN?

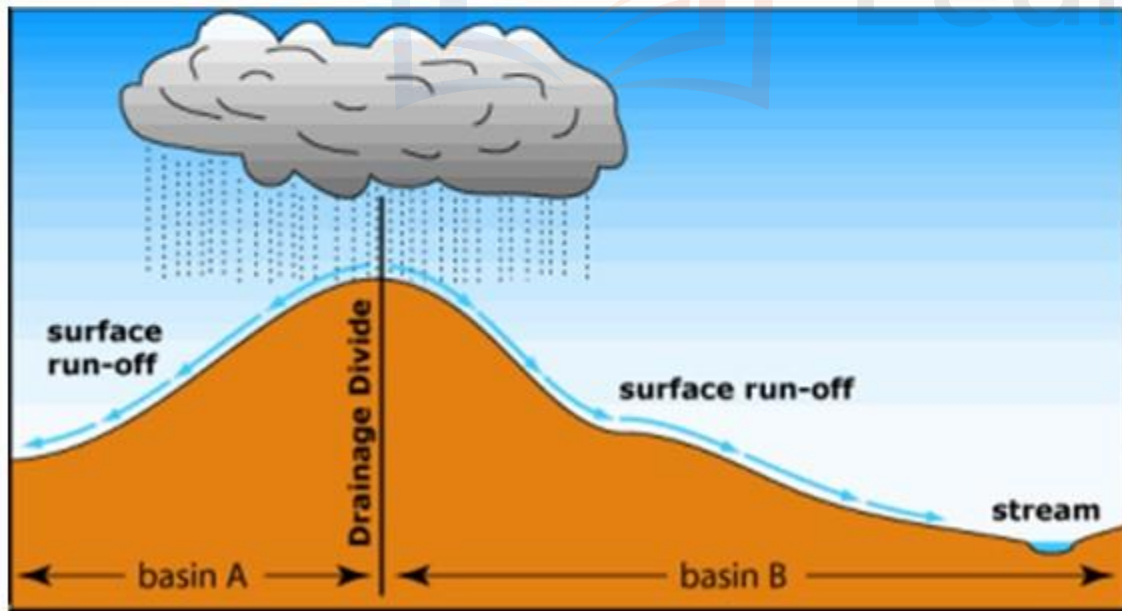


- 1 The total area that contributes water to a **single drainage system** of river is known as the **river's drainage basin**.
- 2 The drainage basin therefore includes both the streams, rivers and land surface.
- 3 It **acts as funnel** by collecting all the water within the area and channeling it to a single point.

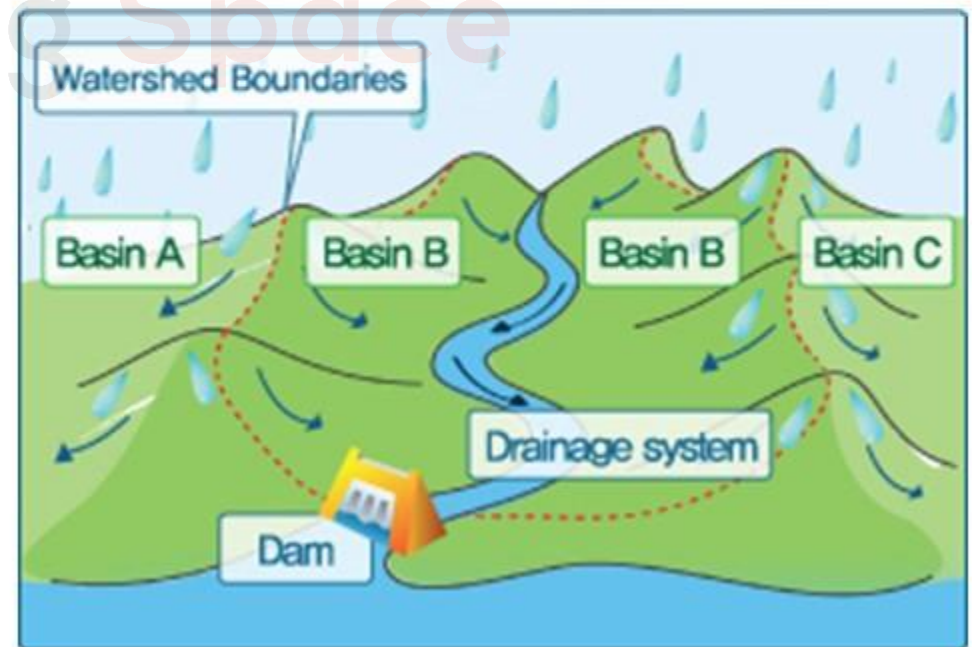
INDIAN DRAINAGE SYSTEM

WHAT DO YOU MEAN BY A THE TERM DRAINAGE DIVIDE?

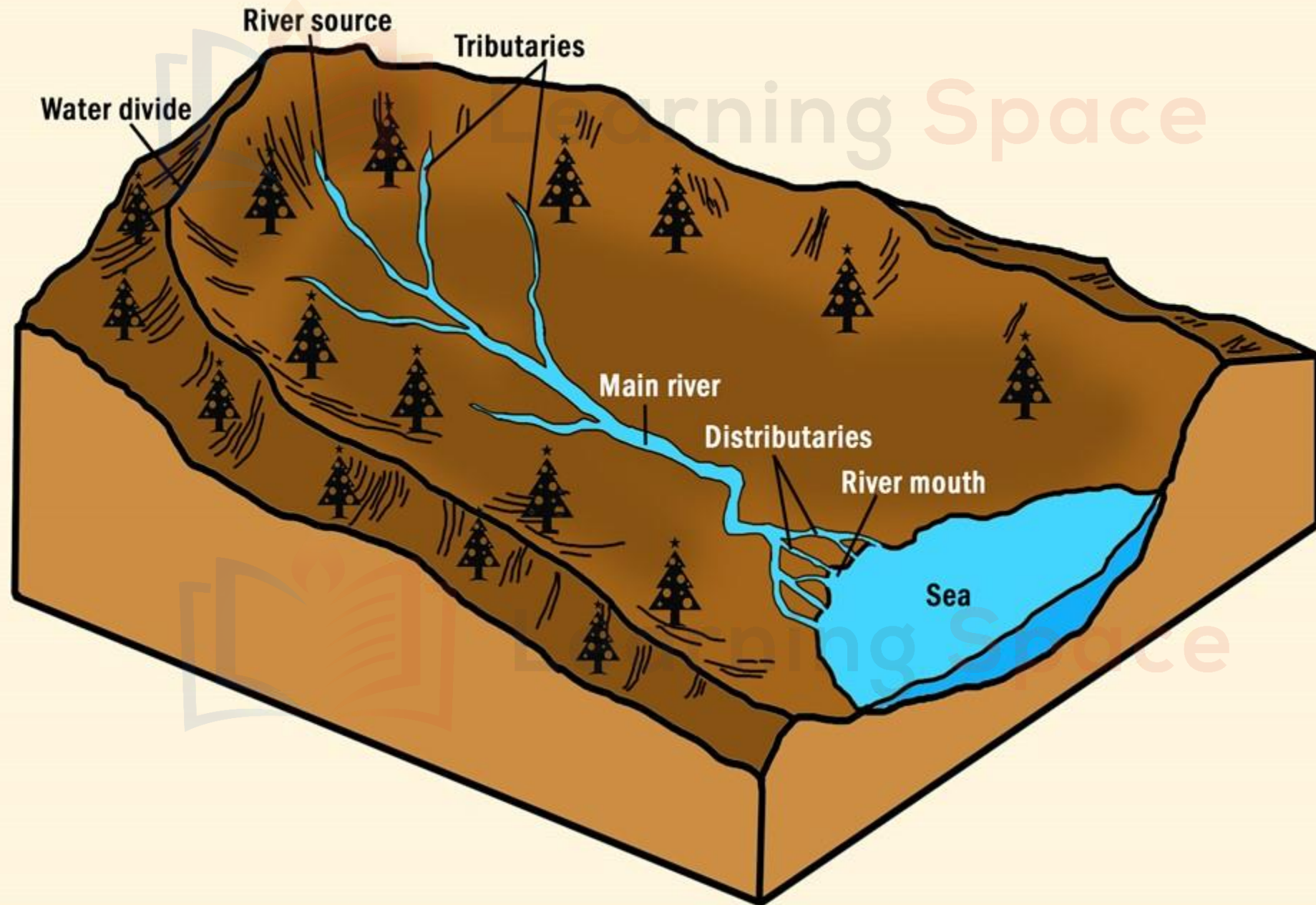
- 1 Adjacent drainage basins are separated from one another by a drainage divide.
- 2 Drainage divide is usually a ridge or a high platform or a mountain system.



- 3 Drainage divide is conspicuous in case of hilly topography like the Himalayas and it is not well marked in plains like Ganga plains.



DRAINAGE BASIN



INDIAN DRAINAGE SYSTEM

WHAT DO YOU MEAN BY DRAINAGE PATTERN?

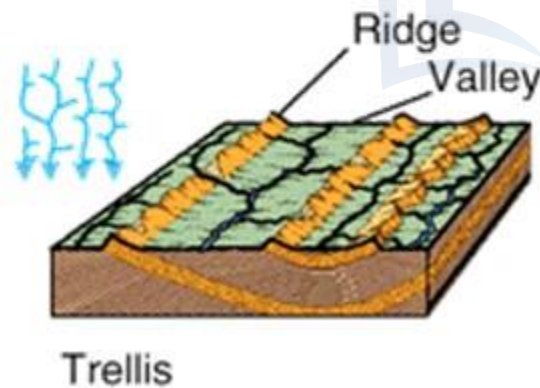
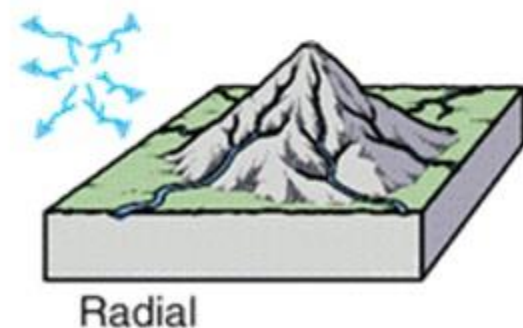
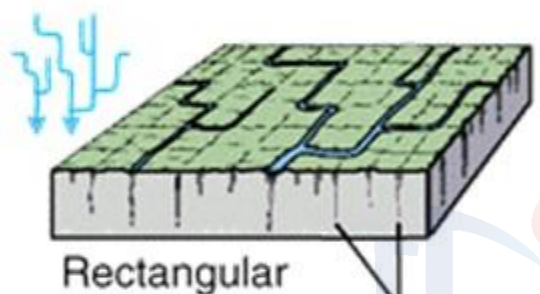
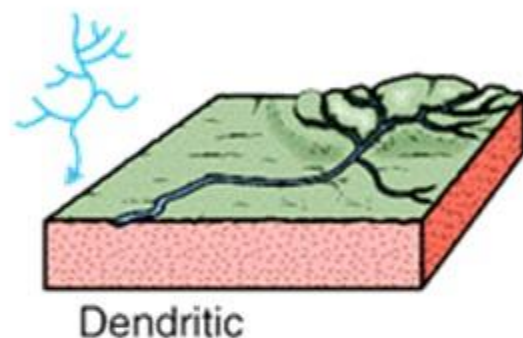


Dendritic drainage of the Yarlung Tsangpo River, in Tibet as seen from space

- 1** Geometric arrangement of streams in region gets determined by
 - **Slope**
 - **Rock resistance** to weathering and erosion
 - **Climate**
 - **Structural controls** of the landscape
- 2** This kind of **geometric arrangement** of the streams is known as a **Drainage Pattern**.
- 3** In other words, it refers to a design, which a river and its tributaries form together, from its source to its mouth.

INDIAN DRAINAGE SYSTEM

WHAT DO YOU MEAN BY DRAINAGE PATTERN?



4

There are different types of drainage patterns:

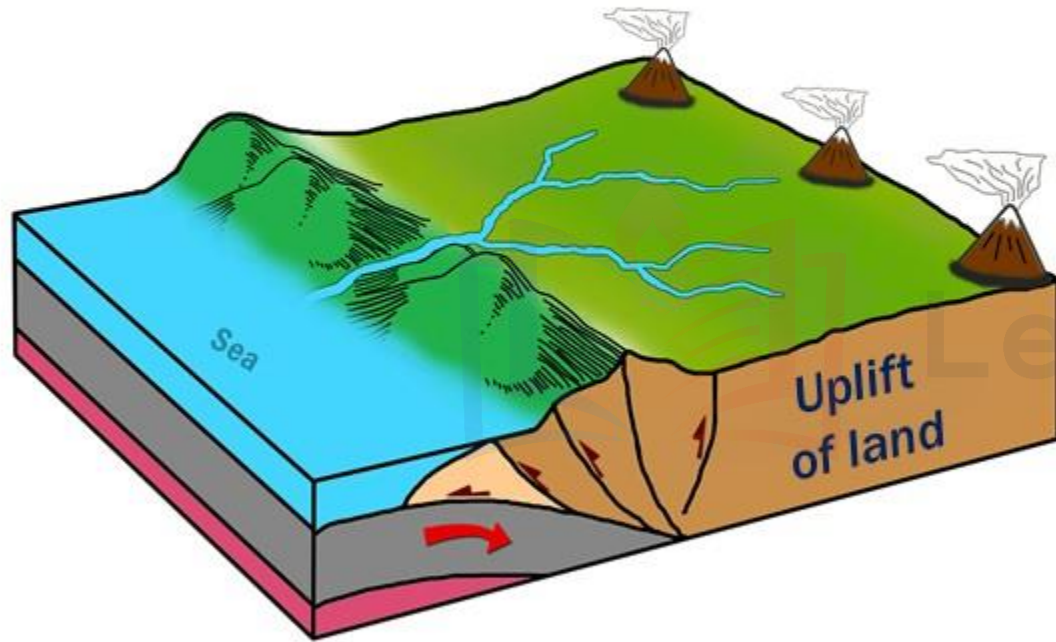
1. The Antecedent or Inconsequent Drainage
2. Consequent Drainage
3. Subsequent Drainage
4. Superimposed Drainage
5. Dendritic Drainage
6. Trellis Drainage
7. Rectangular Drainage
8. Radial Drainage
9. Parallel Drainage
10. Centripetal Drainage

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

1

ANTECEDENT DRAINAGE OR INCONSEQUENT DRAINAGE



*Antecedent drainage cuts through the
tectonic uplift*

1

The rivers that existed before the uplift of the mountains and those that cut their courses southward by making **deep gorges** in the mountains are known as Antecedent Rivers

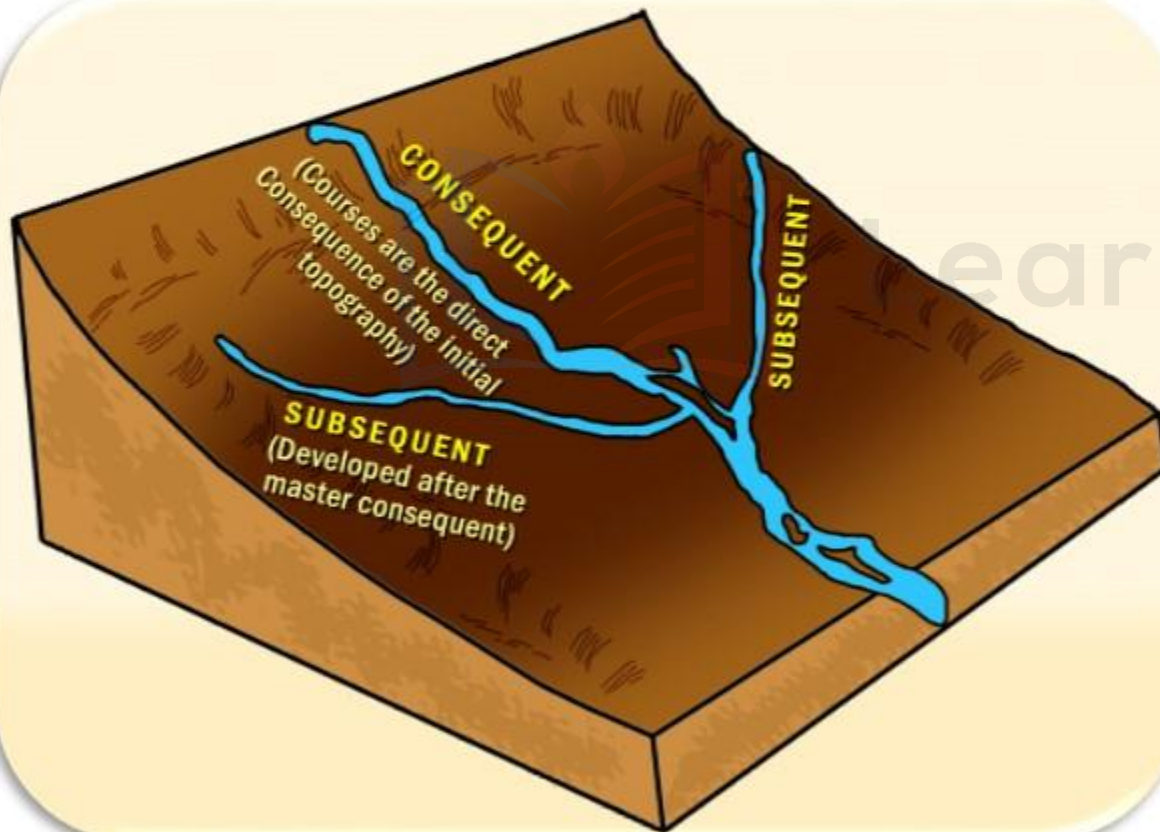
2

Example: the Himalayan rivers like Indus, Sutlej, Ganga Brahmaputra etc

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

2 CONSEQUENT DRAINAGE



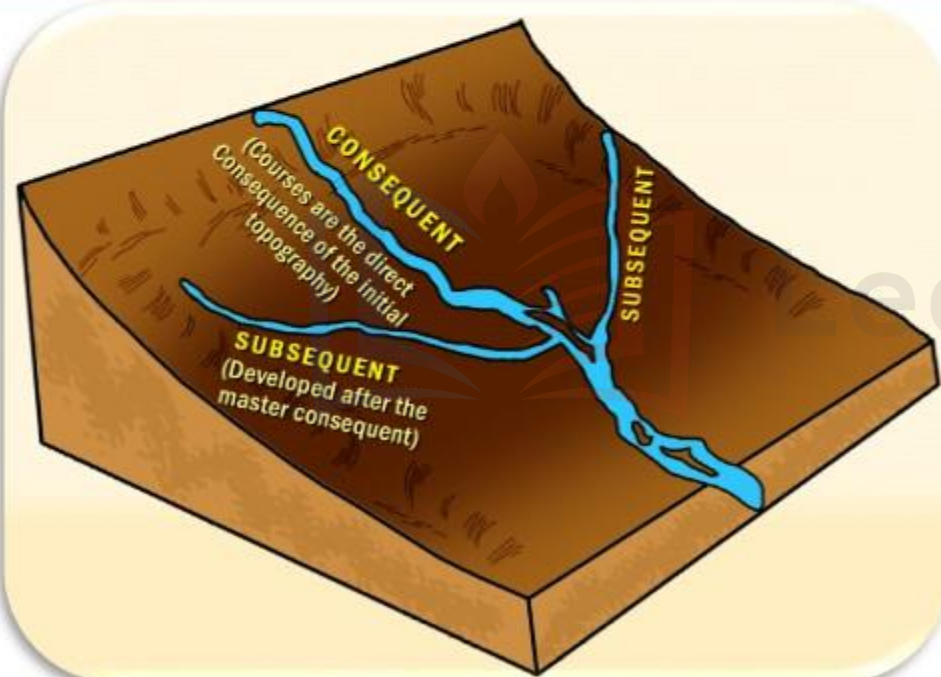
- 1 The rivers which follow the general direction of a slope are known as the Consequent Rivers or Consequent Drainage.
- 2 Most of the rivers of peninsular India are consequent rivers.
- 3 Eg: Godavari, Krishna and Cauvery

INDIAN DRAINAGE SYSTEM

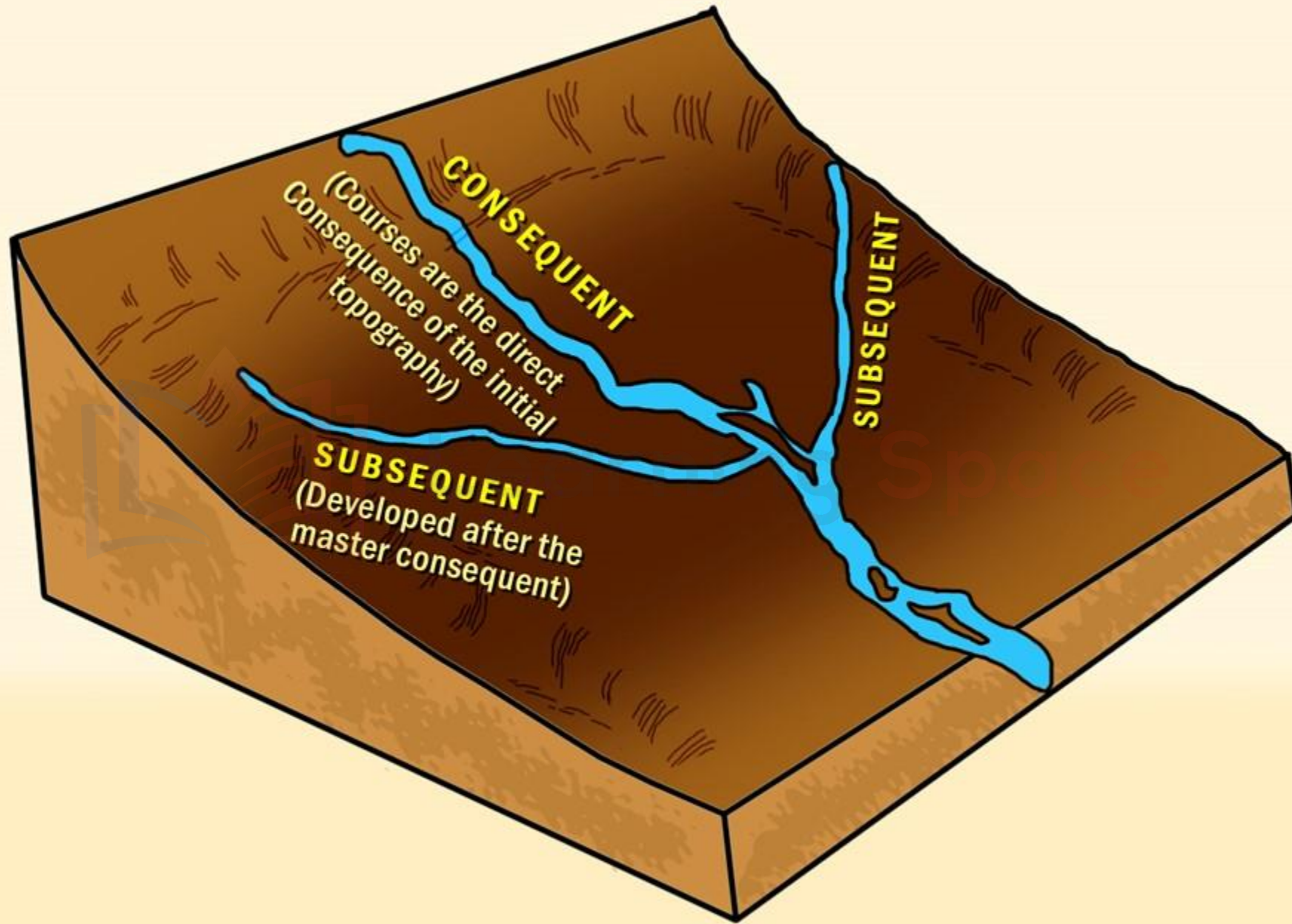
TYPES OF DRAINAGE PATTERN

3 SUBSEQUENT DRAINAGE

- 1 A tributary stream that is eroded along an underlying belt of non-resistant rock after the main drainage pattern (Consequent River) has been established is known as a Subsequent Drainage.



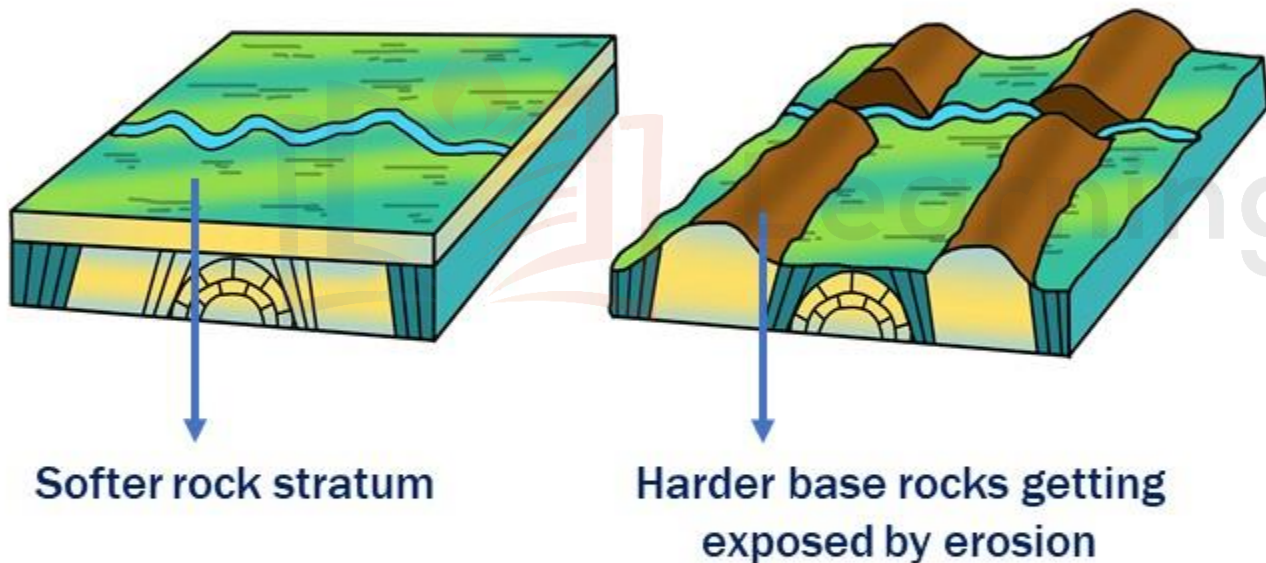
- 2 In simple words, these streams develop after the development of Original Stream.
- 3 Due to the northward slope of the peninsula towards the Great Plains of North India, the rivers originating from the Vindhya and Satpura ranges flow northward into the Ganga system and they are set to be the Subsequent Drainage of the Ganga Drainage system
- 4 Example: Chambal, Sind, Ken, Betwa, and Son



INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

4 SUPERIMPOSED DRAINAGE

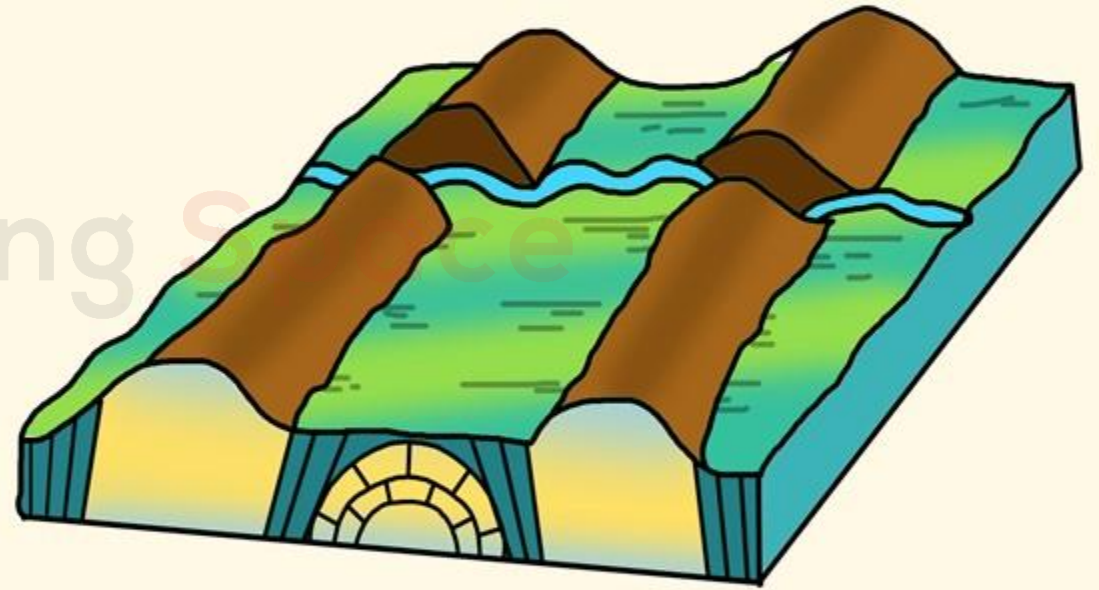
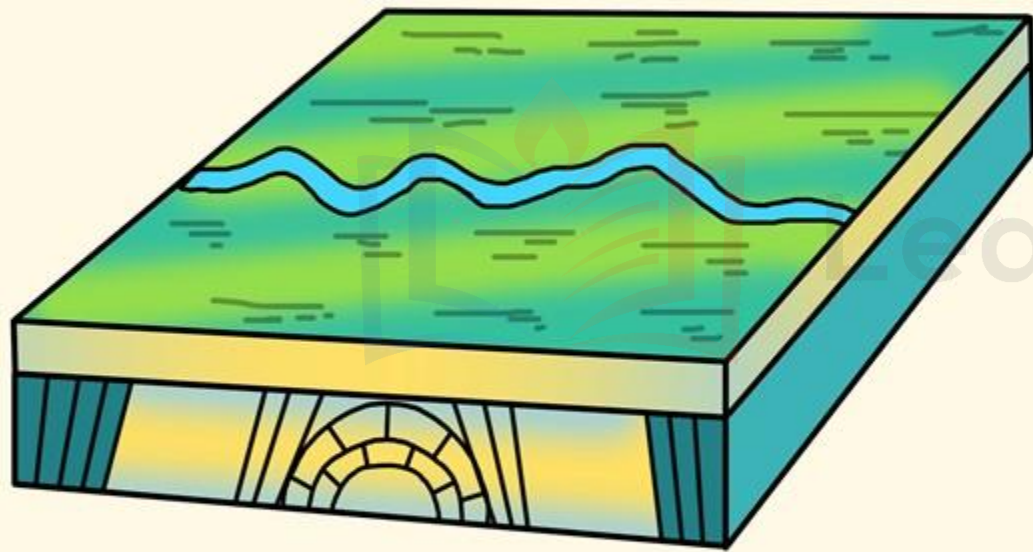


- 1 When a river flowing over softer rock stratum reaches the harder base rocks, but, continue to follow the initial slope, it seems to have no relation with the harder bed rock.

- 2 This type of drainage is called Superimposed Drainage.

- 3 Usually the Drainage patterns are strongly influenced by the rockiness and softness of the rock material over which the river flows.

- 4 However, in the Superimposed Drainage system you get a situation, where the drainage pattern does not correspond to the hardness or the softness of the bed rock.

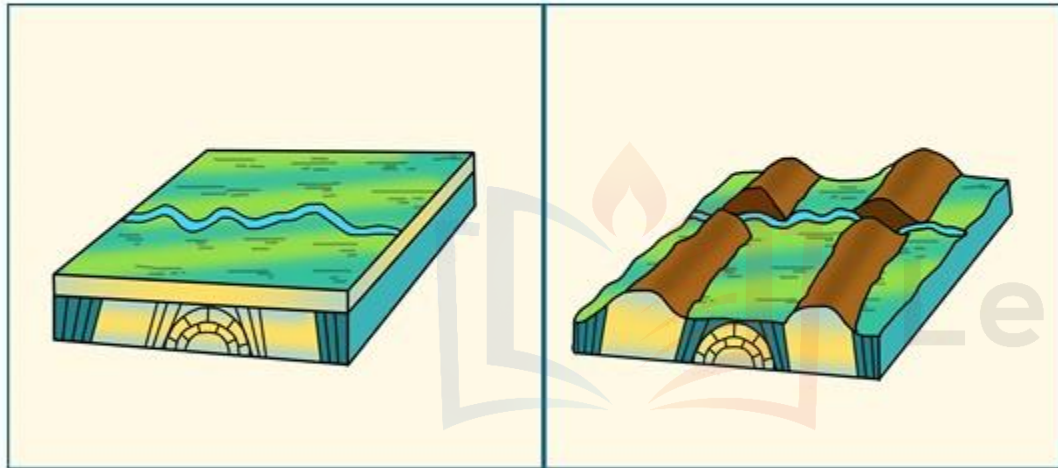


In simple words, the river flow becomes independent of present day topography. It flows in its initial paths without being influenced by changing topography.

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

4 SUPERIMPOSED DRAINAGE



River directions relate to the original rock cover unaltered by the exposed structures

5 In other words, it is a drainage pattern, which exhibits discordance with the underlying rock structure because it originally developed on a cover of rocks that has now completely disappeared due to continuous denudation.

6 Consequently, river directions relate to the original rock cover, which got eroded and the rivers have been able to retain their original courses unaffected by the newly exposed structures.

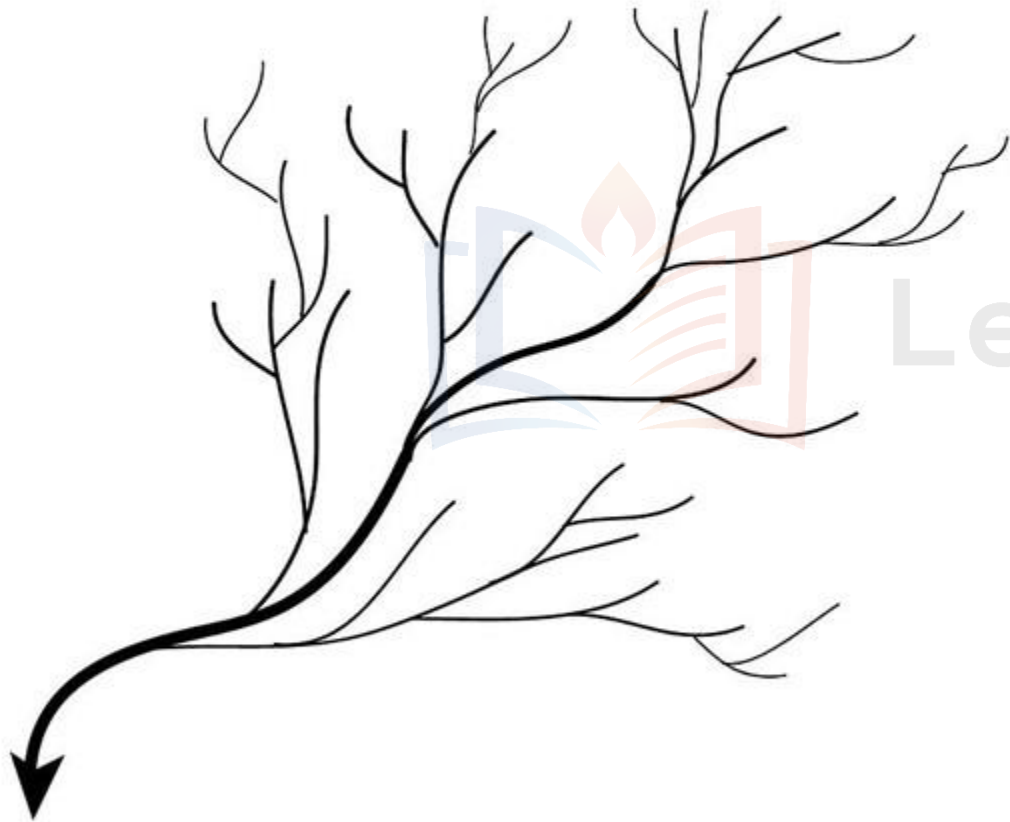
7 Example: Damodar, Subarnarekha

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

5

DENDRITIC DRAINAGE



1

Dendritic means that the drainage system **resembles the appearance of a tree**.

2

This is an irregular tree branch shaped pattern.

3

A dendritic pattern develops in a terrain which has **uniform lithology**.

4

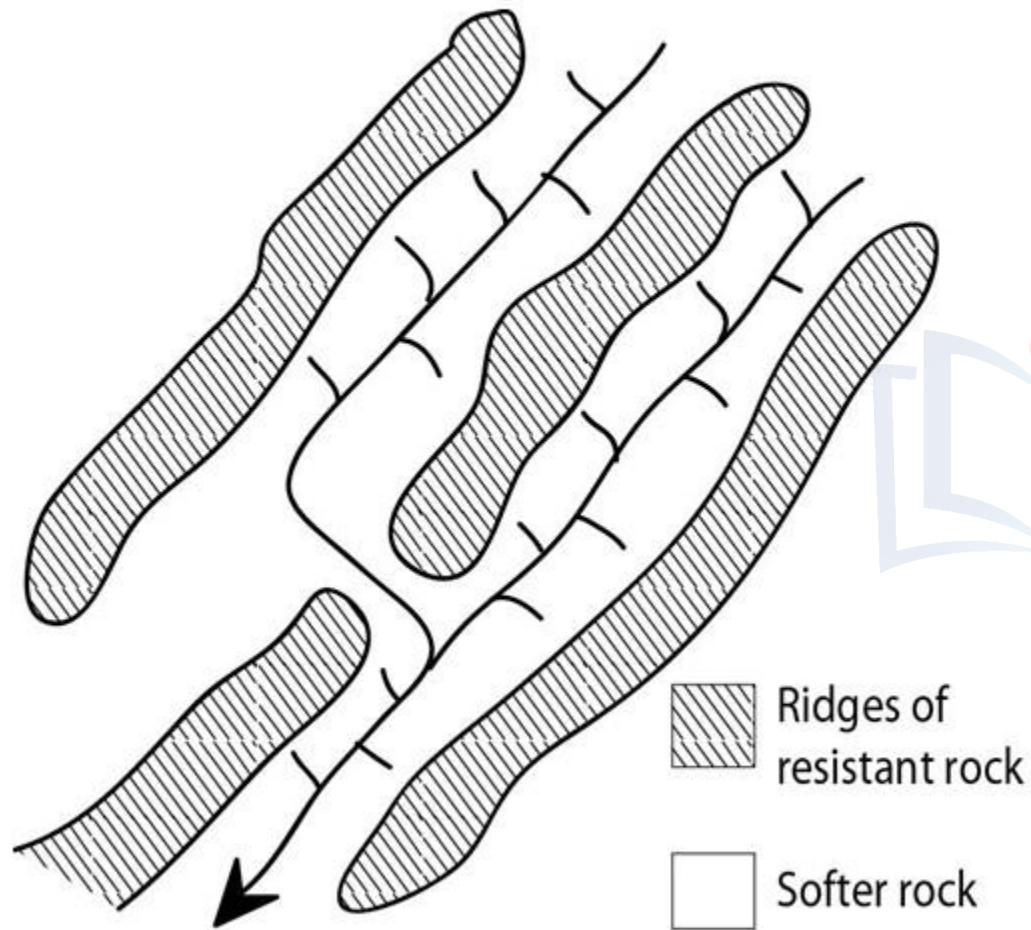
Example: most of the rivers of Indo Gangetic plains.

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

6

TRELLIS DRAINAGE



1 This is characteristic of **folded topography**.

2 This pattern exist in nearly parallel mountains, where drainage patterns are influenced by **rock structures of variable resistance and folded strata**.

3 In this pattern the **short subsequent streams meet the main stream at right angles** or we can say that the tributaries flow roughly parallel to each other.

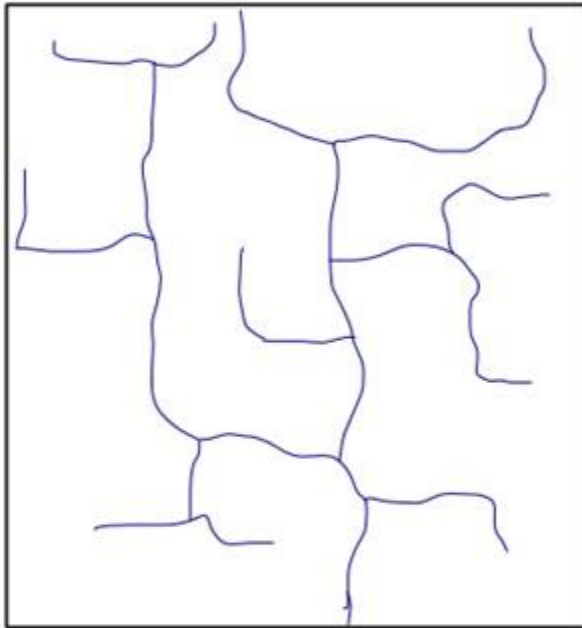
4 Example: **the folded mountains of Singhbhum in Chotanagpur Plateau** have a drainage of Trellis pattern.

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

7

RECTANGULAR DRAINAGE



It differs from the Trellis Drainage Pattern in so far as it is more irregular and its tributaries are neither as long nor as parallel as in Trellis Drainage

1

A rectangular pattern is formed because of faulting and presence of excessive joints in a landscape.

2

This directs the stream in patterns of right angle turns, that is, the main stream bends at right angles and the tributaries join at right angles creating completely rectangular patterns.

3

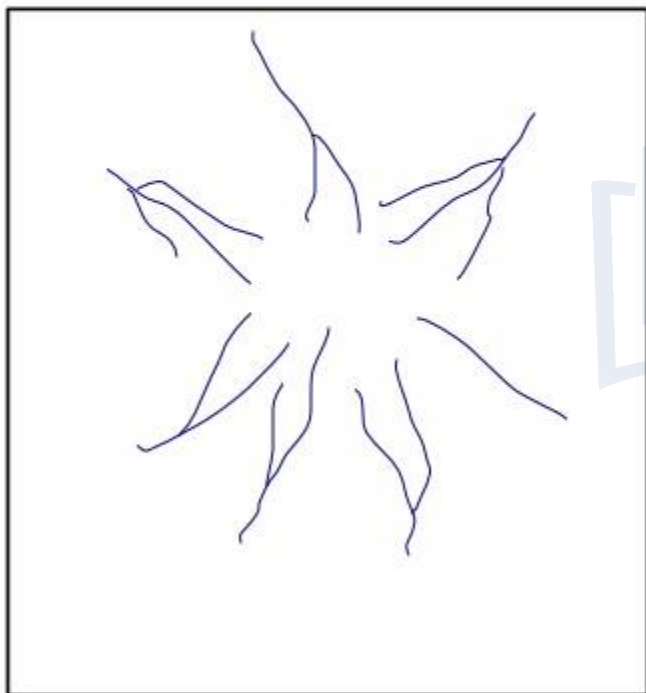
Example: the drainage pattern found in the Vindhyan mountains of India

INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

8

RADIAL DRAINAGE



- 1 Radial pattern of Drainage results from streams flowing off a central peak or a dome.
- 2 It is a pattern characterized by outflowing rivers, which drains down in all directions from a central point.
- 3 Example: rivers originating from the Amarkantak mountain systems.

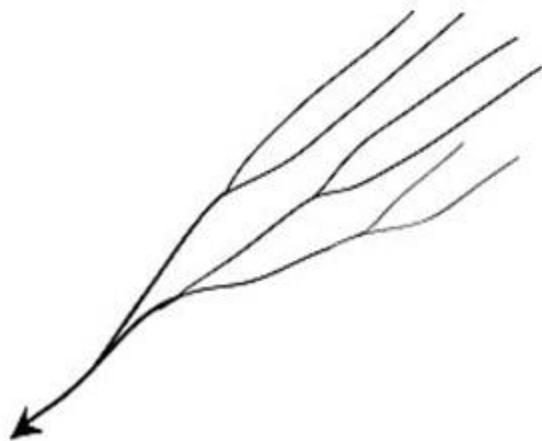
INDIAN DRAINAGE SYSTEM

TYPES OF DRAINAGE PATTERN

9 PARALLEL DRAINAGE

1 The drainage pattern in which the river flows almost parallel to each other is known as Parallel Drainage.

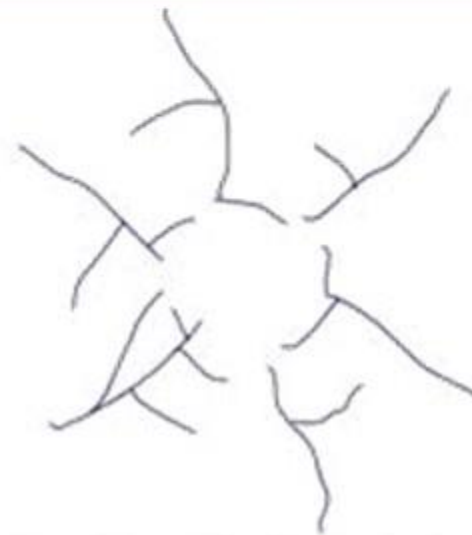
2 Example: The rivers that originate in the Western Ghats like Sharawati in Karnataka, Talpona in Goa, Periyar river in Kerala.



10 CENTRIPETAL DRAINAGE

1 In this patterns the streams converge into a low lying basin from all the sides.

2 Examples: Streams of Ladakh, Tibet and the Baghmati and its tributaries in Nepal



INDIAN DRAINAGE SYSTEM

DRAINAGE SYSTEMS OF INDIA



- 1 Indian drainage system consists of a large number of small and big rivers.
- 2 It is the outcome of the evolutionary process of the three major physiographic units and the nature and characteristics of precipitation.

INDIAN DRAINAGE SYSTEM

THE HIMALAYAN DRAINAGE



Rivers fed by snow and precipitation

- 1 The Himalayan drainage system has evolved through a long geological history.
- 2 It mainly includes the Ganga, the Indus and the Brahmaputra river basins.
- 3 Since these are fed both by melting of snow and precipitation, rivers of this system are perennial.
- 4 These rivers pass through the giant gorges carved out by the erosional activity carried on simultaneously with the uplift of the Himalayas.

INDIAN DRAINAGE SYSTEM

THE HIMALAYAN DRAINAGE

- 5** Besides deep gorges, these rivers also form **V-shaped valleys, rapids and waterfalls** in their mountainous course.

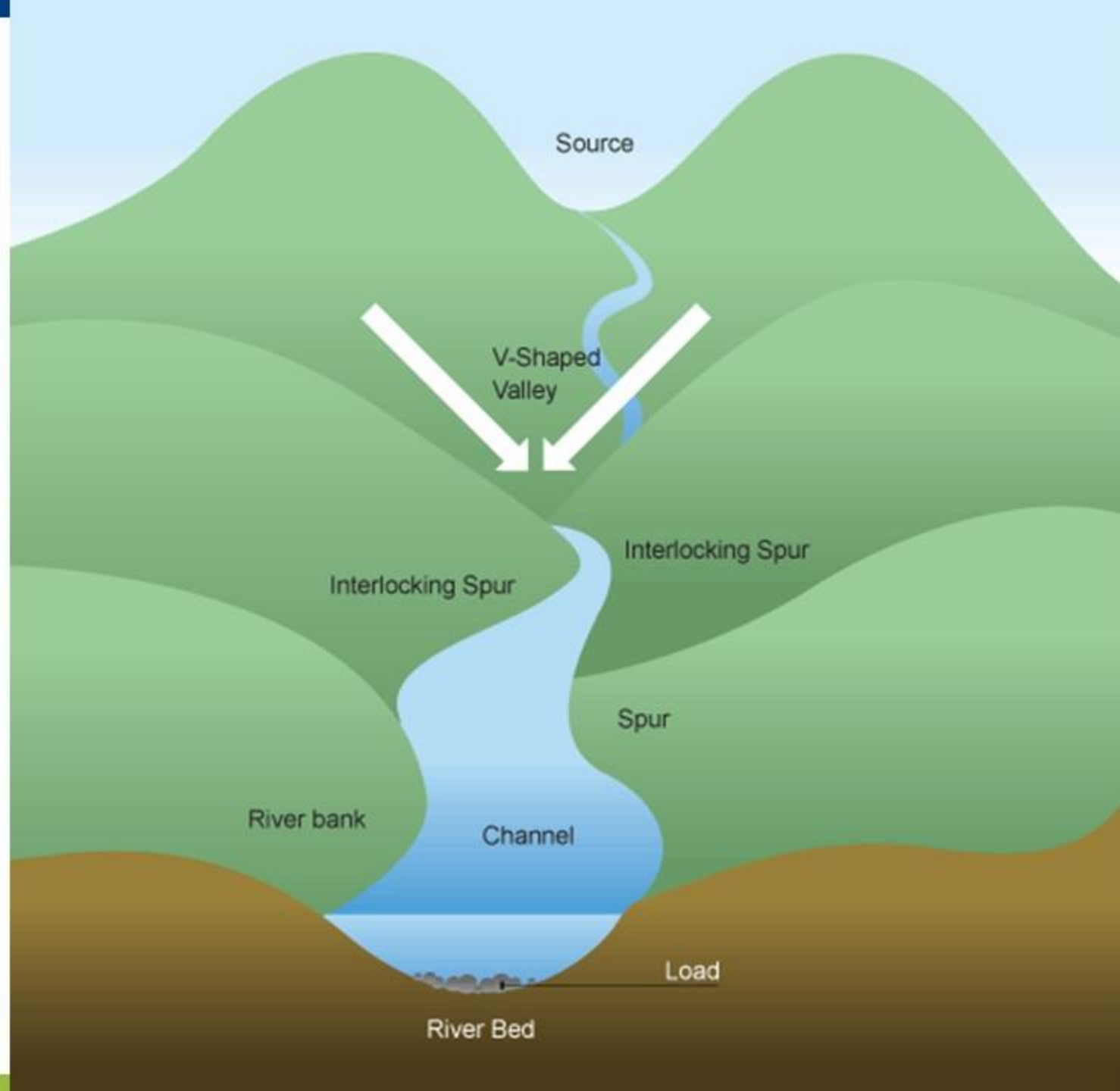


V-shaped valleys

- 6** While entering the plains, they form depositional features like flat valleys, **ox-bow lakes, flood plains, braided channels, and deltas** near the river mouth.



Ox-bow lake formation



INDIAN DRAINAGE SYSTEM

THE HIMALAYAN DRAINAGE

- 7** In the Himalayan reaches, the course of these rivers is **highly tortuous**, but over the plains they display a **strong meandering tendency** and shift their courses frequently.



Flooded North Bihar due to Kosi river

- 8** For example, River Kosi, also known as the '**Sorrow of Bihar**', has been notorious for frequently changing its course.
- 9** The Kosi brings huge quantity of sediments from its upper reaches and deposits it in the plains.
- 10** The course gets blocked, and consequently, the river changes its course.

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